An explicit quartic in 318 variables

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Abstract

We specify an integer quartic in 318 total variables whose rational input projection is the integers under the stated arithmetic premises of the elliptic construction. The local construction uses four factors for sixteen tests. Reciprocal abscissas have even valuations at every good unramified place, allowing smaller powers in those factors. The auxiliary point needs formal depth ten, which still gives comparison precision forty. A common elliptic point and its fixed translate use four field coordinates satisfying three quadrics. Normalizing the numerator logarithms by one common approximation factor avoids a separate denominator-logarithm coordinate. The complete quadratic system, expanded quartic, and independent exact verification certificates accompany the paper. The displayed member has 5060 monomials and coefficient height $80989^{109858}+1$.

1 Families and arithmetic premises

All variable counts include the distinguished input X. For a fixed integer $N \geq 1$ set $A_N = 80989^{2N+1}$. The coefficient tables define 290 primitive integer quadratic polynomials and their sum of squares

$$P_N = \sum_{j=1}^{290} Q_{N,j}^2$$

in 318 coordinates. The parameters N, A_N are fixed coefficients, not variables.

Theorem 1.1. *Assume the arithmetic premises specified below. For every $N \geq 27464$ and every $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^{317}\ P_N(x, \mathbf{y}) = 0.$$

Thus $A_{27464} = 80989^{54929}$ specifies a defining quartic. Every P_N with $N \geq 1$ has degree exactly four, content one, and 5060 nonzero coordinate monomials. For every $N \geq 23$,

$$H(P_N) = 80989^{4N+2} + 1.$$

The explicit member has coefficient height $80989^{109858}+1$, whose binary length is 1791283.

The following arithmetic inputs have precisely the scope stated in [1, 2, 3]. Set

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad D = 11 - 6\rho = (3-\rho)^2, \qquad E : y^2 = x^3 - Dx,$$

$$P = 2(1+\rho, 2\rho) = \left(\frac{57-28\rho}{8}, \frac{-660+417\rho}{32}\right).$$

We retain the group assertion $2E(K) = \mathbb{Z}P$ and $h(x([n]P)) \geq n^2 - 6$ for nonzero integer n; the five-contact height bound with exponent nineteen and its stated exceptional primes; the pole obstruction outside $2, 3, 17$ and its simultaneous realization for inputs with square reduced denominator, the specified local conditions at those three primes, and real value in $(0, 17)$; and the simultaneous prime-pattern and conjugate reduction-order assertions, including

finite congruence preparation and real approximation. The quantitative selection and fixed denominator dilation used in the construction follow from the analytic estimates as detailed in [5]. These finite coefficient calculations do not prove those arithmetic inputs.

We use the effective sum-height theorem of [4], a proved height consequence with explicit constants. No contact-height amplification or new prime-distribution premise is used in this paper.

The scalar normalizer, root-point charts, comparison weights, and local factors are specified by the coefficient tables. Their arithmetic interpretation agrees with [2, 3]. We give the common point chart and normalized logarithm equations explicitly below, and prove their projection properties together with the local valuation, formal-logarithm, and completeness arguments. Source hashes and exact comparisons of every retained row accompany the tables.

2 A quadratic chart for the common reference point

Write $a_P = x(P)$, $b_P = y(P)$, and set

$$A_P = a_P^2 + D, \qquad B_P = a_P^2 - D, \qquad C_P = D^2 + 6a_P^2 D - 3a_P^4.$$

The constants a_P, b_P, D, A_P, C_P are nonzero. The first four nonvanishing assertions follow from the displayed coordinates and $D = (3-\rho)^2 > 0$ at both real embeddings; exact arithmetic gives

$$C_P = -\frac{94897379}{4096} + \frac{4195821}{256}\rho, \qquad \mathrm{Nm}(C_P) = -\frac{8203357319351}{16777216} \neq 0.$$

Retain four field coordinates S_H, T_H, f, F_0. Put

$$\begin{aligned} L_f &= 2b_P D(a_P T_H - b_P), \\ Q_f &= a_P A_P S_H T_H + b_P A_P S_H - b_P B_P T_H + 4a_P^2 D, \\ L_g &= 2b_P D(a_P S_H + b_P), \\ Q_g &= -a_P A_P S_H T_H - b_P B_P S_H + b_P A_P T_H - 4a_P^2 D. \end{aligned}$$

Impose the three quadratic equations

$$\begin{aligned} L_f f &= Q_f, \\ L_g F_0 &= Q_g, \\ D^2 f F_0 + a_P D(f + F_0) + a_P S_H T_H + b_P(S_H - T_H) + D &= 0. \end{aligned} \tag{1}$$

Lemma 2.1. *These equations describe exactly the points $H \in E(K)$ for which H and $H - P$ are finite with nonzero abscissas, through*

$$S_H = \frac{y(H)}{x(H)}, \qquad T_H = \frac{y(H-P)}{x(H-P)}, \qquad f = \frac{1}{x(H)}, \qquad F_0 = \frac{1}{x(H-P)}.$$

In particular $fF_0 \neq 0$ at every solution. The statement includes all finite exceptional cases of the displayed equations.

Proof. For the algebraic calculation abbreviate $a = a_P$, $b = b_P$, $S = S_H$, $T = T_H$ and $g = F_0$. Thus $b^2 = a(a^2 - D)$. Define

$$\Phi = aS^2T^2 - (a^2 - D)(S^2 + T^2) + 2(a^2 + D)ST + 4aD$$

and the polynomial expressions

$$\begin{aligned} 2bDf_0 &= -(a^2 + D)S + (a^2 - D)T - bS^2 - aS^2T, \\ 2bDg_0 &= (a^2 + D)T - (a^2 - D)S - bT^2 + aST^2. \end{aligned}$$

Direct expansion gives

$$\begin{aligned} Df_0^2 + S^2 f_0 - 1 &= \frac{a^2S^2 - b^2}{4ab^2D}\Phi, \\ Dg_0^2 + T^2 g_0 - 1 &= \frac{a^2T^2 - b^2}{4ab^2D}\Phi, \\ L_f f_0 - Q_f = -a\Phi, &\qquad L_g g_0 - Q_g = a\Phi. \end{aligned} \tag{2}$$

Let $r_f = L_f f - Q_f$, $r_g = L_g g - Q_g$, and $R = D^2 fg + aD(f+g) + aST + b(S-T) + D$. The further exact identity

$$\begin{aligned} L_f L_g R = D^2 r_f r_g &+ (aDL_g + D^2 Q_g) r_f \\ &+ (aDL_f + D^2 Q_f) r_g - aD^2 C_P \Phi \end{aligned} \tag{3}$$

shows that every solution has $\Phi = 0$, including the cases in which one leading coefficient vanishes. Also

$$(2bD)^2 R(f_0, g_0) = D^2\big((a^2 - D) - aST - b(S-T)\big)\Phi.$$

If $L_f L_g \neq 0$, these identities give $f = f_0, g = g_0$. If $L_f = 0$, then $T = b/a$ and $\Phi = 0$ forces

$$S = \frac{a^4 - 6Da^2 + D^2}{2b(a^2 + D)}, \qquad L_g = -\frac{aDC_P}{a^2 + D} \neq 0.$$

Thus $g = g_0 = 1/a$. The coefficient of f in R is $D(a^2 + D)/a \neq 0$, so $f = f_0$. The case $L_g = 0$ is symmetric. In particular both leading coefficients cannot vanish simultaneously; at $S = -b/a, T = b/a$ one has $Q_f = C_P \neq 0$.

Equation (2) now proves $fg \neq 0$ and gives the points $H = (f^{-1}, S/f)$ and $H' = (g^{-1}, T/g)$ on E. Two more identities prove the translation:

$$a(f_0 T + g_0 S) + b(g_0 - f_0) = S + T,$$

$$f_0 g_0 (aS + b)^2 - a(1 - af_0)^2 = -\frac{(aS + b)^2\big(aST + b(S-T) + D\big)}{4b^2D^2}\Phi.$$

If $x(H) \neq a$, the second identity gives the abscissa of $H - P$ from the chord slope $(x(H)S + b)/(x(H) - a)$. The first states that $H, -H', -P$ are collinear, and therefore gives the required ordinate. If $x(H) = a$, then $S = \pm b/a$. At $S = b/a$ the formula gives $f = -a/D \neq 1/a$, which is impossible. At $S = -b/a$ it gives $H = -P$ and

$$T = -\frac{a^4 - 6Da^2 + D^2}{2b(a^2 + D)}, \qquad g = \frac{4b^2}{(a^2 + D)^2},$$

which are the coordinates of $H' = -2P$ in the stated form. Thus $H' = H - P$ in all cases.

Conversely, the chord-and-tangent formulas for any allowed H give $\Phi = 0$ and the expressions f_0, g_0 above; the same explicit doubling computation covers $H = -P$. The displayed identities then prove all three equations. This proves both directions. □

3 Even valuations and four local factors

Let v be a place of K above an unramified prime $p > 76176$, normalized by $v(p) = 1$. In particular $v(D) = 0$. Write $f(Q) = 1/x(Q)$ when defined, and put $f(O) = 0$, with $v(0) = +\infty$.

Lemma 3.1. *If $Q \in E(K_v)$ and $x(Q) \neq 0$, then every finite value of $v(f(Q))$ is even.*

Proof. Put $a = v(x(Q))$. If $a < 0$, then $v(x^3) < v(Dx)$, so $2v(y) = 3a$ and a is even. If $a > 0$, then $v(Dx) < v(x^3)$, so $2v(y) = a$ and again a is even. For $a = 0$ the assertion is immediate. If $y = 0$, the nonzero abscissas are $\pm(3 - \rho)$, both units, so this case is included. □

The root points have nonzero integer indices

$$Q_1 = [n]P, \quad Q_2 = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P.$$

Write $B = \max(|n|, |a|, |t|, |b|)$ and let

$$\mathcal{H} = 2^{110}210^{1800}\left(\prod_{p\le 76176} p\right)^{19} + 128.$$

This is the explicit constant in the retained contact-height estimate [3]. The auxiliary points and their exact group differences are

$$U_0 = H, \quad V_0 = H - P, \quad U_i = [u_i]P, \quad V_i = U_i - Q_i \quad (1 \le i \le 4).$$

For $i > 0$ the charts allow either U_i or V_i to equal O. The denominator points H, V_0 are excluded from the exceptional classes in Section 4 before their ratios are used. Let σ be conjugation and define the tests

$$A_i = f(U_i), \qquad B_i = \sigma f(V_i), \qquad D_H = f(H),$$
$$C_i = L_i - L_i', \qquad I = L_1 - \beta, \qquad \beta = 51X^2 + 2504,$$
$$G = L_3^2 - L_2(L_2 - 2)L_3L_4 + (L_2 - 2)^2L_4^2 - 1.$$

The intended orders are one for the nine A_i, B_i, twenty for D_H, ten for C_1, I, and forty for C_2, C_3, C_4, G.

For each nonzero square scale a_* the detector retains

$$t_* = a_*\lambda, \quad \delta = t_* - 1, \quad F(t_*) = \prod_{\theta\in\Theta}(t_* - \theta), \qquad \Theta = \{0, -1/2, -1, 1/2399, -32/407\},$$

$$F(t_*)e = (a_*\delta)^5, \qquad \delta^3 q = e. \tag{4}$$

On $\delta \neq 0$ these give

$$R = \frac{F(t_*)}{a_*^5\delta^6}, \qquad q = \frac{1}{R\delta^4}.$$

At a contact prime p, δ is a unit and $\nu = v_p(R)$ is odd, so $v(q) = -\nu$ at every place above p.

Use the four factors

$$T_1 = 1 + q^{79}C_2^2 + q^{38}D_H^2 + q^4A_1^4 + q^8B_0^8, \tag{5}$$
$$T_2 = 1 + q^{79}C_3^2 + q^{38}C_1^4 + q^4A_2^4 + q^8B_2^8, \tag{6}$$
$$T_3 = 1 + q^{79}C_4^2 + q^{38}I^4 + q^4A_3^4 + q^8B_3^8, \tag{7}$$
$$T_4 = 1 + q^{79}G^2 + q^2A_4^2 + q^4B_1^4 + q^8B_4^8, \tag{8}$$

and let $W = T_1T_2T_3T_4$.

Lemma 3.2. *At a contact place, the four finite nonconstant valuations in each T_j have exact powers of two divisors $1, 2, 4, 8$, respectively. Consequently $T_j \neq 0$, $v(T_j) \le 0$, and $v(T_j) = 0$ implies $T_j \equiv 1$ in the residue field.*

Proof. In the first column of (5)–(8) the valuation is $2v(C) - 79\nu$, which is odd. In the second column it is one of

$$2\big(v(D_H) - 19\nu\big), \qquad 2\big(2v(C) - 19\nu\big), \qquad 2\big(v(A_4) - \nu\big).$$

The parenthesis is odd, by Lemma 3.1 when needed. The third and fourth column valuations are $4(v(f) - \nu)$ and $8(v(f) - \nu)$, with even $v(f)$. They therefore have the asserted exact divisors. The constant term has valuation zero, distinct from every nonconstant valuation. Thus the minimum is attained exactly once, proving all assertions. A zero test removes its term and does not affect the argument. □

Proposition 3.3. *At a contact prime, suppose the retained cleared trace equations and pole system hold. Then $\nu > 0$, and there is one place above that prime at which all sixteen tests have their intended orders.*

Proof. Write $W = w_0 + w_1\rho$. The literal trace equations are

$$w_0^2 - 2w_0T_0 - 2w_1^2 = 0, \qquad e\delta\mathfrak{b} = T_0^2, \qquad e\delta = R^{-1}. \tag{9}$$

Lemma 3.2 shows that $W \neq 0$. If $w_0 = 0$, the first row would imply $w_1 = 0$, a contradiction. Hence

$$T_0 = \frac{\mathrm{Nm}(W)}{2w_0} = \frac{1}{\mathrm{Tr}(W^{-1})} \neq 0, \qquad \mathfrak{b} = \frac{R}{\mathrm{Tr}(W^{-1})^2} \neq 0.$$

Put $\ell = \max_{v|p} v(W) \leq 0$. The trace inequality gives $v_p(\mathrm{Tr}(W^{-1})) \geq -\ell$ and therefore

$$v_p(\mathfrak{b}) \leq \nu + 2\ell.$$

This valuation is odd. The pole obstruction forces $0 < v_p(\mathfrak{b}) \leq \nu + 2\ell$, whence $\nu > 0$ and $\ell > -\nu/2$. At a place attaining ℓ, each factor has valuation at least ℓ, because all four factor valuations are nonpositive and sum to ℓ. Each nonzero term in that factor therefore has valuation strictly greater than $-\nu/2$.

For the first column this gives $v(C) > (79/2 - 1/4)\nu \geq 39.25$, hence $v(C) \geq 40$. The second-column C_1, I terms give $v(C) > (38/4 - 1/8)\nu \geq 9.375$, hence order ten. For D_H the inequality is

$$v(D_H) > (19 - 1/4)\nu \geq 18.75.$$

Its valuation is even, so it is at least twenty. The A_4 inequality gives $v(A_4) > 3\nu/4 \geq 3/4$; those for the third and fourth columns give respectively $v(f) > 7\nu/8$ and $v(f) > 15\nu/16$. Their even integer valuations are therefore at least two, proving all order-one requirements. Zero tests satisfy every required order. □

Degenerate detector parameters. The cleared identities (4) have exactly the domain properties proved in [2, Lemma 2.1]. Since $a_* \neq 0$ and $1 \notin \Theta$, $F(t_*) = 0$ is impossible. If $q = 0$, then $e = 0$ and $\delta = 0$. At $t_* = 1$ the contact radical is one, because all prime divisors of the numbers $1 - \theta$ lie below the fixed cutoff. Thus no contact argument uses a reciprocal on its zero stratum.

4 Formal comparisons at depth ten

This section proves the depth implication without presupposing integrality of the ratios. It includes all allowed common points and identity numerators. Lemma 2.1 gives

$$Df^2 + S_H^2 f = 1, \qquad H = (f^{-1}, S_H/f).$$

The group calculation in the elliptic source writes $H = [m]R_0 + T$, where $P = 2R_0$, $m \in \mathbb{Z}$, and $T \in E[2](K)$. Put $h = m/2$ for its rational free index; this notation does not assert a half-integer multiplication on the curve.

4.1 Denominator exclusions

Every point of $E[2](K)$ or $\pm P + E[2](K)$ which is permitted by the finite chart has reciprocal abscissa a unit at all primes above $p > 76176$. This follows from the explicit translation identities and norm values

$$\mathrm{Nm}(x(P)) = 41^2/64, \quad \mathrm{Nm}(x(P) - (3-\rho)) = 17^2/64, \quad \mathrm{Nm}(x(P) + (3-\rho)) = 63^2/64$$

given in the source. If H were such a point, Proposition 3.3 would contradict any contact prime. All comparison radicals would therefore be one. The retained height inequality for the root labels, obtained from the effective sum-height theorem and the contact-height estimate, then gives

$$B^2 \le 5\mathcal{H}.$$

The unchanged scalar certificate gives $|a| > (40\mathcal{H})^{10}$ for $N \ge 27464$ [3, Section 4]. Since $B \ge |a|$, these bounds are incompatible. This argument uses no quotient of logarithms. Consequently $m \notin \{0, 2\}$, and H and $V_0 = H - P$ are nontorsion.

In the common point chart, $T_H = 0$ would make $H - P$ a nonzero two-torsion point. The preceding argument therefore proves $T_H \neq 0$ for every solution at $N \ge 27464$. This conclusion uses only the reciprocal-abscissa tests and the effective height estimates; it is independent of the normalized logarithm equations below.

The denominator logarithm has the following global nonvanishing certificate. In particular its rational approximant has numerator with no root modulo the split prime 71 and denominator with no root modulo the split prime 7. Their leading coefficients are units at those primes. A root in K would give an integral local root and a forbidden residue root. Thus the rational approximant never vanishes on K. The common point chart gives $z(V_0) = -1/T_H \neq 0$, since V_0 is nontorsion. Consequently $z(V_0)\mathcal{R}(Df(V_0)^2)$, the algebraic approximate logarithm used in the mathematical interpretation, is nonzero. No coordinate for this product is needed in the quadratic system. The analytic formal logarithm is treated separately below by the formal logarithm isometry. All unchanged chart denominators are justified by the retained source equations.

4.2 Formal logarithm errors

Write $z = -x/y$. At a nonidentity formal point of depth $r = v(z) \ge 1$, one has $v(f) = 2r$ and

$$\log_E(Q) = z(Q)\mathcal{A}(Q), \qquad \mathcal{A}(Q) = G(Df(Q)^2), \qquad G(w) = \sum_{j \ge 0} g_j w^j,$$

where

$$g_0 = 1, \quad g_1 = -2/5, \quad g_j = \frac{4j-5}{4j+1} g_{j-1} \quad (j \ge 2).$$

This is the formal logarithm expansion established in the elliptic source; it also follows by integrating the invariant differential. Set $\mathcal{A}(O) = 1$.

The two approximation functions are

$$G_2(w) = 1 - \frac{2}{5}w - \frac{2}{15}w^2, \qquad \mathcal{R}(w) = \frac{N(w)}{Q(w)}, \qquad Q(w) = 1 - \frac{1526}{1073}w + \frac{513}{1073}w^2,$$

$$N(w) = 1 - \frac{9776}{5365}w + \frac{2941}{3219}w^2 - \frac{1024}{13949}w^3 - \frac{1920}{237133}w^4 - \frac{1024}{711399}w^5 - \frac{1408}{5928325}w^6 + \frac{2432}{103152855}w^8.$$

The denominator Q has no root modulo 7, and the numerator N has no root modulo 71. Both primes split in K; their leading coefficients are units at the respective primes. Thus neither polynomial has a root in K. These finite residue assertions are included in the independent exact checker.

Lemma 4.1. *At every place in question, $\mathcal{A}(Q) \in 1 + p^{4r}\mathcal{O}$ for a nonidentity formal point of depth r. The relative errors of the retained quadratic Taylor polynomial G_2 and rational approximant $\mathcal{R}$ are at least $12r$ and $40r$, respectively. At O the errors are zero.*

Proof. The recursion gives $v(g_j) \geq -(4j+1)/(p-1)$, since each denominator divides a product of factors of $(4j+1)!$. When $4j+1 < p$, it is integral. Since $v(Df^2) = 4r$, the nonconstant terms have valuation at least $4r$: this is immediate in the integral range, and for the remaining $j \geq 19044$ follows from $4rj - (4j+1)/(p-1) \geq 4r$. The same argument starting at $j = 3$ or $j = 10$ gives the Taylor tail bounds $12r$ or $40r$. For the retained rational approximant, the exact identity

$$Q(w)G_9(w) - N(w) = \frac{2970726}{254443709}w^{10} - \frac{41697666}{6361092725}w^{11}$$

is part of its coefficient certificate. All denominators in the approximant are supported at primes at most 37, and $Q(0) = 1$; therefore $\mathcal{R} - G_9$ has valuation at least $40r$. Division by the unit $\mathcal{A}(Q)$ preserves these bounds. At O the analytic unit quotients are defined by their constant value one, so no expression $\log_E(O)/z(O)$ is used. □

4.3 Quadratic equations for the alternate logarithm ratios

For the numerator-point charts write

$$z(V_i) = -\frac{\mathcal{Q}_i}{\mathcal{K}_i}, \qquad \mathcal{K}_i \neq 0 \quad (1 \leq i \leq 4).$$

These identities include $V_i = O$, where $\mathcal{Q}_i = 0$. Let

$$\mathcal{B}_0 = \mathcal{R}(Df(V_0)^2), \qquad \mathcal{B}_1 = G_2(Df(V_1)^2), \qquad \mathcal{B}_i = \mathcal{R}(Df(V_i)^2) \quad (2 \leq i \leq 4).$$

At the identity these functions take the value one. Every $\mathcal{B}_i$ is an affine expression in retained coordinates of its approximation circuit, and $\mathcal{B}_0 \neq 0$ by the preceding residue certificate.

For each i introduce one field coordinate Z_i. Let $\widetilde{L}_i$ denote the stored, unconjugated logarithm ratio, and put $L'_i = \sigma(\widetilde{L}_i)$ in the comparison tests. Impose

$$\mathcal{B}_0 Z_i = \mathcal{Q}_i \mathcal{B}_i, \qquad \widetilde{L}_i \mathcal{K}_i = Z_i T_H. \tag{10}$$

These are quadratic equations over K, hence four rational quadratic equations per numerator point. Since $\mathcal{B}_0\mathcal{K}_i \neq 0$, they give uniquely

$$Z_i = \frac{\mathcal{Q}_i\mathcal{B}_i}{\mathcal{B}_0}, \qquad \widetilde{L}_i = \frac{\mathcal{Q}_i\mathcal{B}_i T_H}{\mathcal{K}_i\mathcal{B}_0}.$$

The previously established exclusion $T_H \neq 0$ therefore yields

$$\widetilde{L}_i = \frac{z(V_i)\mathcal{B}_i}{z(V_0)\mathcal{B}_0}, \qquad L'_i = \sigma\left(\frac{z(V_i)\mathcal{B}_i}{z(V_0)\mathcal{B}_0}\right).$$

This equality holds also when $V_i = O$, with both sides equal to zero. Conversely, the intended elliptic points and approximate ratios give unique values of the Z_i satisfying (10). Thus these equations have exactly the required projection on the arithmetically admissible domain. Their only additional potential boundary $T_H = 0$ has already been excluded without division by a logarithm; it introduces no extra rational solutions at the defining coefficient threshold.

4.4 Integrality before precision

Fix the place supplied by Proposition 3.3. The positive f valuations put H, U_i on the primary formal branch, and V_0, V_i on the conjugate formal branch. In particular $v(z(H)) \geq 10$. The group identities

$$[m]U_i = [2u_i]H, \qquad [m-2]V_i = [2(u_i - n_i)]V_0$$

kill the two-torsion component of H. Applying the formal logarithm at the respective places gives the full ratios

$$r_i = u_i/h, \qquad r_i' = (u_i - n_i)/(h-1).$$

The denominators are nonzero by the preceding exclusions, and the formal logarithms of H, V_0 are nonzero by the formal logarithm isometry at these places [7, Theorem IV.6.4].

The primary approximate ratio is exactly

$$L_i = r_i \frac{\mathcal{A}(H)}{\mathcal{A}(U_i)}.$$

The alternate ratio is r_i' times a conjugated quotient of units $G_2/\mathcal{A}$ or $\mathcal{R}/\mathcal{A}$, with the denominator using $\mathcal{R}$ at V_0. This remains true when a numerator point is O, since both the approximate ratio and its rational index ratio then equal zero.

At the odd prime p, $h, h-1 \in \mathbb{Z}_{(p)}$ differ by one, so at least one is a unit. The corresponding full ratio is integral because the integer labels are integral. Multiplication by the indicated analytic units preserves its valuation. The positive-order test $C_i = L_i - L_i'$ then transfers integrality to the other approximate ratio and hence to both full ratios. This step uses only the initially known formal branches and the unit assertion in Lemma 4.1; it does not use the deeper precision conclusion.

Now $L_i = z(U_i)/z(H)$ is integral. If $U_i \neq O$, it follows that $v(z(U_i)) \geq v(z(H)) \geq 10$. The primary relative error is therefore at least forty. If $U_i = O$, its error is exactly zero. The alternate relative error is at least twelve for $i = 1$ and forty for $i > 1$. The full ratios are integral, so these relative errors give absolute congruences. Together with the cross tests, they yield

$$L_i \equiv r_i \equiv r_i' \pmod{p^{k_i}}, \qquad k_1 = 10, \quad k_i = 40 \ (i > 1).$$

Multiplying by the integral numbers h and $h-1$ and subtracting shows

$$L_1 \equiv n \pmod{p^{10}}, \qquad L_i \equiv n_i \pmod{p^{40}} \quad (i > 1). \tag{11}$$

At an inert prime conjugation preserves the local valuation, and the identical argument applies. This proves the required local comparison without an assumption that either numerator is nonzero or that h is an integer.

5 Global soundness and completeness

Equation (11), the order-ten input test, and the order-forty Lucas test give precisely the divisibility statements used by the retained global height proof. The Lucas polynomial has integer coefficients and all three relevant approximate ratios are integral at the extracted place, so substitution of the integer indices changes its value by a multiple of p^{40}. The input test similarly gives the original order-ten divisibility for the numerator of $n - \beta$. Thus the same height contradictions force the Lucas identity and then $n = \beta$. Since 51 is squarefree, $51x^2 + 2504 \in \mathbb{Z}$ implies $x \in \mathbb{Z}$: a reduced denominator v would satisfy $v^2 \mid 51$. The scalar lower bound and every numerical threshold in the explicit construction are unchanged. This proves soundness at the scope stated in Theorem 1.1.

For completeness retain the source's choice of integer root labels, scalar point, scales, fixed local preparation, contact primes, and common auxiliary group index. At each selected primary

place its construction gives depth at least twenty for H, formal depth at least one for each other relevant point, cross precision at least forty for C_2, C_3, C_4, input cross precision at least twelve, and exact target and Lucas relations to the needed precision. Consequently every nonconstant term in (5)–(8) has positive valuation when $\nu = 1$. For example the lower bounds for the first factor are $1, 42, 4, 8$. Hence $W \equiv 1$ at the designated place. At the opposite place each factor has valuation at most zero and is one modulo the prime if its valuation is zero. Thus the trace of W^{-1} reduces to 1 or 2 and is a unit, as required for the retained pole input.

At the fixed exceptional primes, take neighborhoods of the already prepared points on which all test valuations are bounded below. For $\lambda = u/(D_0^2 w^2)$, choose the fixed integer D_0 sufficiently divisible before choosing the primes. Since the zero contact makes u a selected prime outside the fixed exceptional set, the argument of [2, Section 3] gives

$$v(q) = 2v(a_*) - 3v(\lambda) \geq 2v(a_*) + 6v(D_0).$$

All exponents of q in the new factors are positive, so the new terms are arbitrarily small simultaneously on those neighborhoods. This preserves the local squareclasses and lower valuation bounds at $2, 3, 17$ and the scalar conditions.

At both real embeddings, q tends to zero through positive values and the test values remain bounded on the selected real neighborhoods. Every test power in the four factors is even. Thus $W \to 1$ through positive values and the pole input has the same positive decay as in the retained construction. Outside the selected primes the detector has even valuation; the squared trace denominator therefore preserves the square reduced-denominator condition. The pole realization gives all remaining witnesses. This proves completeness without strengthening the simultaneous prime-pattern input.

6 Quadratic specification and exact verification

The scalar weight circuit is

$$q_2 = q^2, \quad q_4 = q_2^2, \quad cq_4 = q, \quad q_8 = q_4^2, \quad cq_{19} = q_8^2, \quad q_{38} = q_{19}^2, \quad cq_{79} = q_{38}^2.$$

At a contact these force the displayed powers, and their degenerate stratum is already covered by (4). The required squared tests are retained coordinates or affine combinations of them. The fourth powers of B_1 and the eighth powers of B_0, B_2, B_3, B_4 are already present in the logarithm circuits. Exactly five additional field values are needed:

$$C_1^4,\ I^4,\ A_1^4,\ A_2^4,\ A_3^4.$$

Each is computed by two rational quadratic equations from its existing square, using $(u+v\rho)^2 = u^2 + 2v^2 + 2uv\rho$.

For each detector the four factors cost eight rational coordinates. Two intermediate field products cost four more; the final field product uses the retained coordinates of W. Thus each detector costs twelve new coordinates and fourteen quadratic equations. The complete count is

retained coordinates, including input	272
five shared field powers	10
four factors and intermediate products for three detectors	36
total coordinates	318
retained quadratic rows	238
shared power rows	10
factor and product rows	42
total quadratic rows	290

The exact scripts use rational arithmetic, clear each row's coefficient denominators, and divide by its integer content. Since a sum of rational squares vanishes precisely when each term vanishes, the resulting quartic has the required input projection. The exact specification is generated by `build_quartic318_v18.py` and verified by `check_quartic318_v18.py`. The checker imports no producer code. It independently reconstructs all twenty-two modified rational rows, checks all 268 retained rows against the hash-pinned coefficient table, and expands the entire sum of squares. It verifies coordinate occurrence, degree, content, coefficient support, the height comparison, and the residue certificates for the logarithm approximation. The independent general chart certificate verifies the polynomial identities in the proof of Lemma 2.1. The unchanged local-factor certificate checks 70912 supplementary valuation cases; those cases supplement the valuation proofs above.

For the six rational components of (1), the stored primitive integer rows are an invertible rational linear combination of the primitive components in the displayed order. The transformation matrix is included in the construction report; its exact determinant is

$$-\frac{1}{362963666515968}.$$

Each resulting row is again divided by its integer content. Hence this choice of rows has precisely the same rational zero set. The independent checker verifies the determinant and each transformed row exactly.

The unique coefficient involving A_N^2 is the coefficient of the actual scalar contact product squared, equal to A_N^2+1. Every other coefficient is a nonzero constant or a nonzero constant multiple of A_N. This proves support independence for positive A_N. The maximum constant coefficient C has 1520 binary digits, and the maximum linear multiplier is $L = 2\cdot 80989^2 = 13118436242$. Exact integer comparison gives

$$C > L^2, \qquad 80989^{90}+1 < C < 80989^{94}+1, \qquad 80989^{47} > L.$$

Therefore $H(P_N) = \max(C, A_N^2+1)$ for every positive integer N, and $H(P_N) = A_N^2+1$ exactly when $N \geq 23$. This algebraic height threshold does not alter the defining threshold $N \geq 27464$ in the explicit theorem.

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